

Rank Before You Merge: Training-Free Machine-Centric Visual Token Compression for Efficient Vision-Language Model Inference

Zhengxing Yan¹, Hao Sun², Chen Guo³, and Jianpeng Hu^{1,*}

¹Shanghai University of Engineering Science

²Nanjing University of Science and Technology

³East China Normal University

super_fw@163.com; sh1076849939@163.com

guochenjkhjk@qq.com; mr@sues.edu.cn

*Corresponding author

Abstract

Native spatial mergers already compress visual patches before a vision-language model (VLM) reaches its language model, yet most token selectors rank the merger outputs. We call compression *machine-centric* when its units, budget, and interface follow the deployed model’s native computation graph. Rank-Before-Merge (RBM) is a training-free rule that ranks complete native units before applying the unmodified merger. At 25% retention, RBM outperforms matched post-merger L2 selection on all nine text-dense full-split cells across Qwen3-VL, Qwen2.5-VL, and InternVL3, with gains of 11.0–38.4 percentage points and 235–432 OCRBench points. A merger-aware rank prior also improves FastV-k3 on all four locked Qwen3-VL probes at the same budget, although only TextVQA is significant and pure RBM remains stronger on OCRBench. Rank and token-survival analyses attribute the main effect to merger-induced saliency rewriting. RBM adds no learned parameters and increases measured throughput by 68% on Qwen3-VL at 25% retention.

Introduction

Modern VLMs expose hundreds to tens of thousands of visual tokens to the language model. Shorter sequences reduce attention and key-value-cache cost: on our A40, a 25% visual-token rate lowers Qwen3-VL latency from 0.36 to 0.23 seconds and raises throughput by 68%. The question is which evidence should occupy that limited rate.

Merger-equipped models first compress neighboring patches into native spatial units, then select among the merged tokens. If merging changes saliency order, a later top- k rule can discard an answer-bearing unit.

We call compression *machine-centric* when its unit, budget, positions, stream identity, and interface follow the deployed model’s computation graph. It is a systems contract, not a learned importance model. Rank-Before-Merge (RBM) scores complete 2×2 units at the merger input, keeps a fixed per-image budget, and applies the original merger to the survivors (Fig. 1).

We contribute (1) a formulation that states when representation-dependent top- k decisions before and after merging agree; (2) RBM, a training-free compressor with an explicit native-unit and interface contract, validated on three VLMs and two mergers; (3) RankBridge, a locked probe showing that pre-merger rank complements FastV attention; and (4) rank, edge-energy, and swap controls that connect the accuracy gap to merger-induced saliency rewriting.

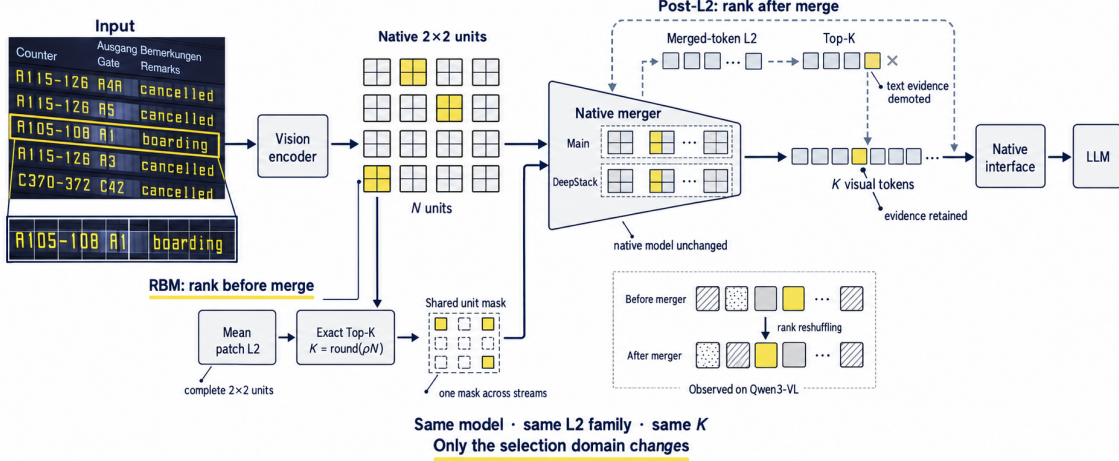


Figure 1: RBM follows the native merge topology. It ranks complete merger-input units, preserves a fixed per-image budget, and passes the selected units through the unmodified merger and language-model interface. The comparison changes selection order, not model weights.

Related Work

Visual-token reduction spans several stages. Methods inside the LLM use query attention or progressive pruning, including FastV, PyramidDrop, FasterVLM, and SparseVLM [1, 2, 3, 4]. Encoder-output selectors use visual [CLS] attention, graph propagation, fitted predictors, or multi-cue suppression [5, 6, 7, 8, 9]. Unified systems combine saliency with diversity or text guidance [10, 11, 12].

Merging changes representation as well as length. ToMe uses bipartite matching; VisionZip preserves dominant tokens and merges the rest; AdaptMerge adds language guidance [13, 14, 15]. TokenPacker learns a compact projector, while GlimpsePrune, VScan, and AnchorPrune target dynamic or Qwen-family compression [16, 17, 18, 19]. Thus, a selector’s input representation is part of the compression rule.

Recent work broadens selection beyond one attention score. QuietPrune and PixelPrune move reduction earlier; TransPrune, V2Drop, and Hi-Lo Prune model token or layerwise variation [20, 21, 22, 23, 24]. Coverage, reconstruction, and functional sensitivity guide ApET, SCoRe, CoIn, ZOO-Prune, and SPaRE [25, 26, 27, 28, 29]. Learned or strongly conditioned methods model semantic labels, information flow, or head importance [30, 31, 32, 33]. Other designs vary token lifetime by layer [34, 35, 36], or preserve discarded evidence through object-centric, local/global, and hybrid aggregation [37, 38, 39]. Document-oriented reduction and provenance audits expose OCR-specific failures [40, 41]. LMMs-Eval and EffiVLM-Bench provide the complementary evaluation layer [42, 43].

RBM isolates the native-merger boundary. Qwen-VL concatenates four neighboring patches before LayerNorm and an MLP; InternVL3 uses pixel-shuffle downsampling [44, 45, 46]. RBM keeps these operators and their interface unchanged but ranks their inputs, enabling a matched pre/post comparison at equal token rate.

Machine-Centric Compression

Selection around a native merger

Let $U = \{u_i\}_{i=1}^N$ be the native merge units and let M be the model’s per-unit merger. A representation-dependent scorer produces $a_i = s_{\text{pre}}(u_i)$ before merging and $b_i = s_{\text{post}}(M(u_i))$ after merging. At budget k , the two decisions are

$$K_{\text{pre}} = \text{TopK}_k\{a_i\}, \quad K_{\text{post}} = \text{TopK}_k\{b_i\}. \quad (1)$$

With deterministic tie breaking, they are equivalent exactly when $K_{\text{pre}} = K_{\text{post}}$. A sufficient condition for agreement at every budget is $b_i = g(a_i)$ for a strictly increasing g . We measure the violation at budget k by

$$D_k = 1 - \frac{|K_{\text{pre}} \cap K_{\text{post}}|}{|K_{\text{pre}} \cup K_{\text{post}}|}. \quad (2)$$

For a fixed set, selection and a per-unit merger are compatible; disagreement arises when the merger changes the ranking that determines the set.

For compression analysis, define the visual-token rate and task distortion as

$$R = \frac{k}{N}, \quad \mathcal{D}(R) = 1 - \frac{S(R)}{S(1)}, \quad (3)$$

where $S(R)$ is the downstream benchmark score at rate R . A lower $\mathcal{D}$ means that more task-relevant information survives. RBM and Post-L2 have identical R , so their vertical separation isolates distortion introduced by the representation used for selection.

Rank-Before-Merge

For every image, RBM groups ViT patch features into the model’s native 2×2 units. For a unit u with input patch vectors $h_{u,1:4}$, it uses the query-blind control score

$$s_{\text{pre}}(u) = \frac{1}{4} \sum_{j=1}^4 \|h_{u,j}\|_2, \quad s_{\text{post}}(u) = \|M(h_{u,1:4})\|_2. \quad (4)$$

It retains $k = \max(1, \text{round}(\kappa N))$ units and applies the native merger. The contract selects complete units, enforces each image’s exact top- k budget, propagates their identities through main and deepstack streams, and leaves the merger, positions, interface, and LLM unchanged. Scoring costs $O(Nd)$ and heap selection $O(N \log k)$. RBM adds no parameters, question tokens, or LLM-attention pass.

Qwen’s merger acts independently per unit, so a fixed kept set produces the same merged representations; this is byte-identical on the verified Qwen3-VL path and separates ranking from forward-operator changes.

Kept visual units (blue) per selector — two OCR cases

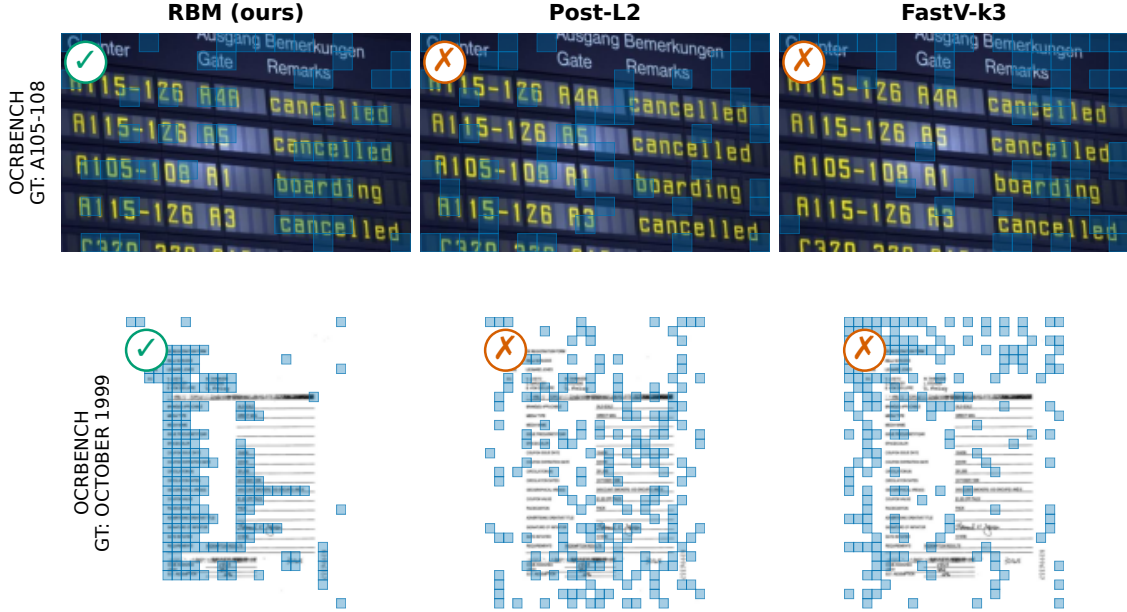


Figure 2: Audited OCRBench examples at 25% retention. RBM, Post-L2, and FastV-k3 preserve different spatial evidence. The examples illustrate selection behavior; aggregate results are reported separately.

A merger-aware prior for FastV

FastV ranks all visual tokens by question-conditioned attention at LLM layer $K = 3$. At $\kappa = 0.25$, RankBridge protects $q = \text{round}(\rho k)$ RBM-ranked positions and fills $k - q$ by FastV rank. A development gate over $\rho \in \{0.1, 0.2, 0.3\}$ selected and locked $\rho = 0.2$ for four $n = 200$ benchmarks. The endpoints are bit-identical to FastV ($\rho = 0$) and RBM in kept-set identity ($\rho = 1$); all candidates remain available until layer K .

Experimental Protocol

We evaluate Qwen3-VL-8B, Qwen2.5-VL-7B, and InternVL3-8B with bf16 weights, greedy decoding, and family-correct native positions. RBM, uncompressed, and Post-L2 use vLLM 0.19 V1 [47]; FastV/RankBridge use HuggingFace eager attention because FastV lacks a vLLM path. The eager native path matches 8/8 samples, and HF/vLLM uncompressed outputs agree on 16/16 held-out cases. Throughput is compared only within vLLM.

We use TextVQA VQA accuracy [48], DocVQA ANLS [49], OCRBench /1000 [50], and GQA normalized exact match [51]; full splits contain 5000, 5349, 1000, and 12578 items. Scorers port lmms-eval [42]. Pre/post tests pair samples and per-image token counts. Nine prespecified text-dense 25% comparisons use 20,000 paired-bootstrap resamples and sign-flip tests with Holm correction (all corrected $p \leq 4.5 \times 10^{-4}$). FastV/RankBridge are secondary paired probes on $n = 200$.

Table 1: Official full-split results. RBM and Post-L2 use the same per-image visual-token count. Δ is RBM minus Post-L2 at 25% retention. OCRBench uses points; other rows use percentage points.

Model	Benchmark (metric)	None	RBM	Post-L2	RBM-Post
Qwen3-VL	TextVQA (VQA acc.)	.844	.605	.222	+38.4
	DocVQA (ANLS)	.956	.481	.238	+24.3
	OCRBench (/1000)	760	547	184	+363
	GQA (EM)	.616	.449	.477	-2.8
Qwen2.5-VL	TextVQA (VQA acc.)	.862	.702	.442	+26.1
	DocVQA (ANLS)	.949	.636	.526	+11.0
	OCRBench (/1000)	817	480	182	+298
	GQA (EM)	.604	.559	.585	-2.6
InternVL3	TextVQA (VQA acc.)	.8338	.7890	.4148	+37.4
	DocVQA (ANLS)	.9221	.7284	.3820	+34.6
	OCRBench (/1000)	852	753	321	+432
	GQA (EM)	.6293	.5993	.6031	-0.4

Compression Results

Matched pre- and post-merger selection

Table 1 isolates selection stage within query-blind L2 scoring. At 25% retention, RBM wins all nine text-dense cells by 11.0–38.4 pp or 235–432 OCRBench points across three VLMs and two merger designs. The gap widens at 12.5% on the reported text-dense cells, although RBM remains lossy relative to no compression.

GQA marks the boundary: Qwen rankings favor Post-L2 by 2.6–2.8 pp, while InternVL3’s -0.4 pp interval crosses zero. A full-split Qwen3-VL control matched at the merger input still gives RBM +27.68 pp TextVQA, +4.59 pp DocVQA, and +235 OCRBench, but -5.64 pp GQA (95% CI $[-6.42, -4.86]$). Selection stage is workload-conditioned.

FastV and the merger-aware prior

At the same 25% budget, RankBridge improves FastV-k3 on all four probes (Table 2). TextVQA gains 3.2 pp ($z = 2.11$); the other 0.5–1.6 pp gains are not significant at $n = 200$. RBM remains 11.6 pp stronger on OCRBench, so the hybrid misses its preset within-1 pp gate. Pre-merger rank adds information to query attention, but one global mixture cannot erase the workload boundary.

On full OCRBench, RBM exceeds FastV by 14.6 pp on Qwen3-VL and 8.7 pp on Qwen2.5-VL; FastV is stronger on Qwen3-VL TextVQA/GQA and Qwen2.5-VL TextVQA. The selectors therefore occupy different regimes.

Figure 3 gives rate-distortion operating points: pre-merger selection lowers text-dense distortion at equal rate, while GQA reverses the ordering.

Table 2: Locked Qwen3-VL probe at 25% retention ($n = 200$). RankBridge protects 20% of its final budget with the pre-merger rank and fills the remainder with FastV-k3 attention.

Benchmark	RBM	FastV-k3	RankBridge	RB+FastV (pp)
TextVQA	.5967	.7633	.7950	+3.2
DocVQA	.4239	.5863	.6023	+1.6
OCRBench	.5801	.4586	.4641	+0.6
GQA	.4150	.5050	.5100	+0.5

Why the Merger Changes Selection

Three Qwen3-VL diagnostics explain the gap (Fig. 4). Pre/post ranks correlate weakly: Spearman is 0.137/0.332/0.360 and top-25% Jaccard is 0.180/0.243/0.278 on DocVQA/TextVQA/GQA. The merger changes the top- k set, not only score scale.

The change is structured. Correlation between post-minus-pre rank shift and Sobel edge energy is +0.439/ +0.155/ +0.036. On DocVQA, RBM-only units have mean edge energy 0.641 versus 0.124 for Post-L2-only units; 92% versus 35% exceed the per-image median. This associates the merger with text-stroke demotion without attributing it to the training objective.

A ranking-swap control holds the post-merger forward path fixed but uses the pre-rank. It exactly selects RBM’s set (Jaccard = 1.000), recovers its accuracy, and yields byte-identical retained outputs. The Qwen3-VL gap therefore comes from ranking, not a different operation on kept units.

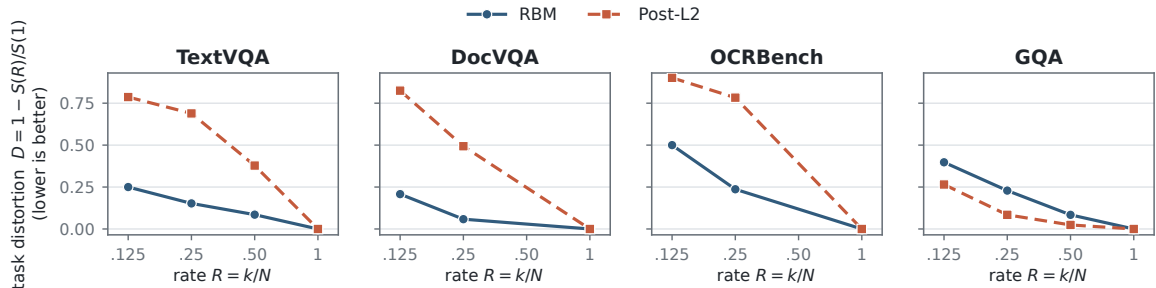


Figure 3: Task rate–distortion operating points on the locked Qwen3-VL diagnostic subset ($n = 200$ per point). Rate is the retained native visual-unit fraction; distortion is the relative loss in official task score from the uncompressed model. Curves connect measured rates only.

What the native merger changes

per-image distributions | 25% retained

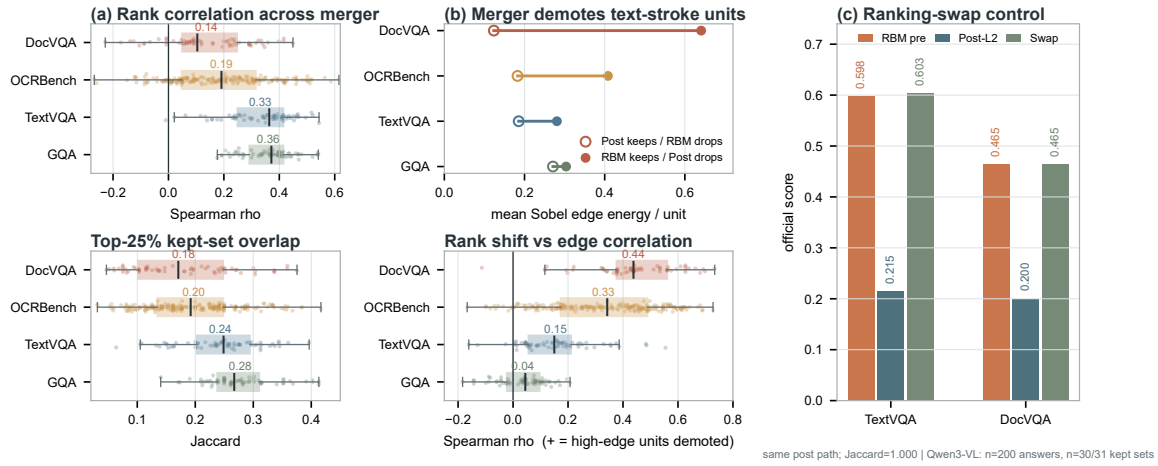


Figure 4: The native merger rewrites selection saliency. Rank agreement is low, text-stroke-rich units are demoted, and a ranking-swap control restores RBM accuracy with the same kept set.

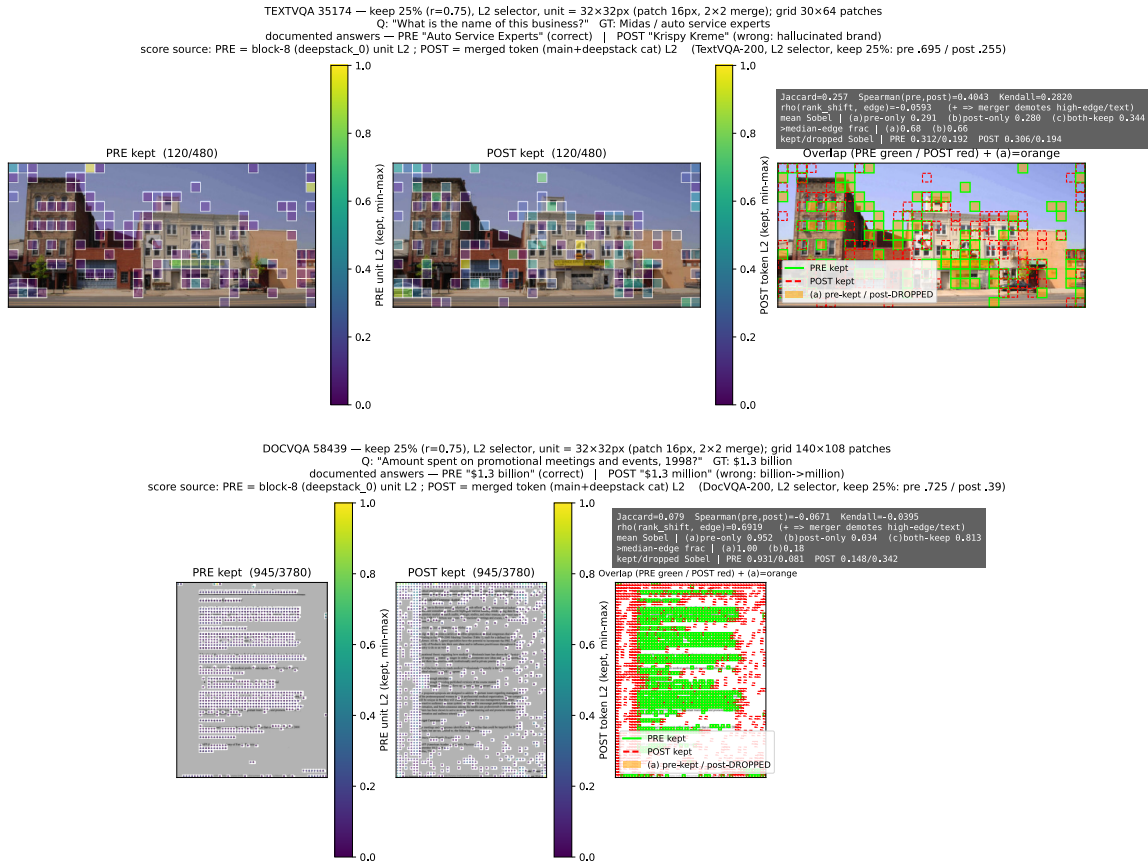


Figure 5: Token-survival maps for audited TextVQA and DocVQA images. Each row shows RBM, Post-L2, and their overlap at the same 25% retention. Green and red outlines identify units kept before and after merging, respectively. These cases illustrate spatial allocation rather than aggregate accuracy.

Efficiency and Scope

On one A40 with vLLM, RBM versus Post-L2 throughput is 4.61 versus 4.52 at 75%, 5.39 versus 5.54 at 50%, and 6.91 versus 6.89 requests/s at 25%; five repeats stay within 3%. RBM gains 12%, 31%, and 68% over no compression; at 25%, latency falls from 0.36 to 0.23 seconds. InternVL3 gains 1.8–2.5 $\times$. RBM leaves ViT work before its scoring tap intact and accelerates the merger and LLM sequences.

Byte-exact attribution holds for Qwen3-VL; Qwen2.5-VL retains an ordering con-found, and InternVL3 provides accuracy replication plus a bounded swap. Models without native merge units are outside scope. FastV/RankBridge use the eager harness, several secondary cells use $n = 200$, and only the TextVQA hybrid gain is significant. Family pixel caps also preclude absolute cross-model comparison. The supported claim is merger-aware, text-dense compression, not a universal selector.

Conclusion

Visual-token selection around a native merger is an operator-order problem. RBM selects complete units at the merger input and applies the unchanged model to survivors. At equal budgets it preserves more text-dense evidence than Post-L2 across three VLMs and two mergers while accelerating inference. Its rank also improves FastV-k3 in four probes, showing complementary signals. Because the merger rewrites saliency and demotes text-stroke units, merger-equipped VLM compression should treat topology and selection stage as first-class constraints.

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